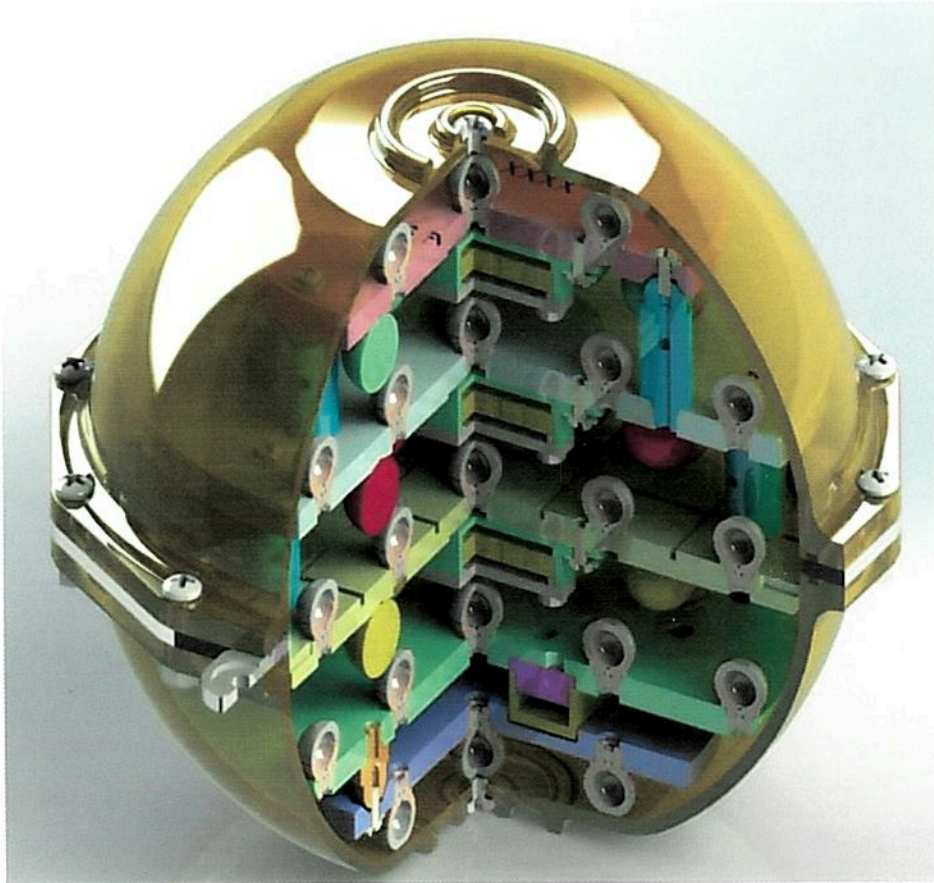




System Standard Model 130

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Product Manual

I. INTRODUCTION

The QalibreMD System Standard was developed by a joint effort between NIST and ISMRM and modeled after the System Standard designed for the Alzheimer's Disease Neuroimaging Initiative (ADNI) MRI core. The spherical diameter of this System Standard was chosen to mimic a human head and to reduce susceptibility distortions.

II. SYSTEM STANDARD COMPONENTS

The System Standard consists of a deionized-water filled spherical shell with a 200 mm inner diameter. Inside the spherical shell is a framework consisting of 5 plates rigidly connected with positioning rods. The plates support 57 fiducial spheres, a 14 element T_1 array, a 14 element T_2 array, a 14 element proton density array, 3 resolution insets (1 mm to 0.6 mm $\pm$ 0.015 mm) and (0.8mm to 0.4mm $\pm$ 0.015 mm), and 2 wedges at 10° for slice profiling. The plates, as shown below, contain the following structures:

Plate 5: 4 fiducial spheres, serial numbers, System Standard ID, T_1 array (blue-green), resolution inset

Plate 4: 13 fiducial spheres, T_2 array (red), 2 resolution insets (1 mm to 0.06mm and 0.8 mm to 0.4 mm)

Plate 3: 21 fiducial spheres, proton density array (yellow),

Plate 2: 13 fiducial spheres, slice profile wedges

Plate 1: 5 fiducial spheres (blue)

The fiducial spheres are 10.0 ± 0.1 mm inner diameters set on a 3 dimensional 40 ± 0.1 mm grid. The origin of the System Standard is defined as the center of the central fiducial sphere as shown in the figure on page 5. The orientation was chosen so that the plates are in the coronal plane with left, right superior, inferior marked on each plate. There is a notch in the mounting plate to allow easy recognition and placement of the System Standard within the scanner and as a holder for the custom level. The System Standard is placed in the same fashion as a patient laying supine on the imaging bed.

1. SPHERE LOCATIONS

Sphere locations are shown in the images below. Note that since the spheres are attached to the bottom, the view is from the bottom of the plate showing a counterclockwise orientation of the spheres from 1-10 on the outer circle and 11-14 on the inner circle. When viewed from the top of the System Standard the orientation of the spheres will be clockwise. *Coronal slices viewed from anterior to posterior will show a clockwise geometry of the spheres.*

Note: All spheres are the same size. The images are sized with respect to the plate size.

Plate 5, T1 Spheres:

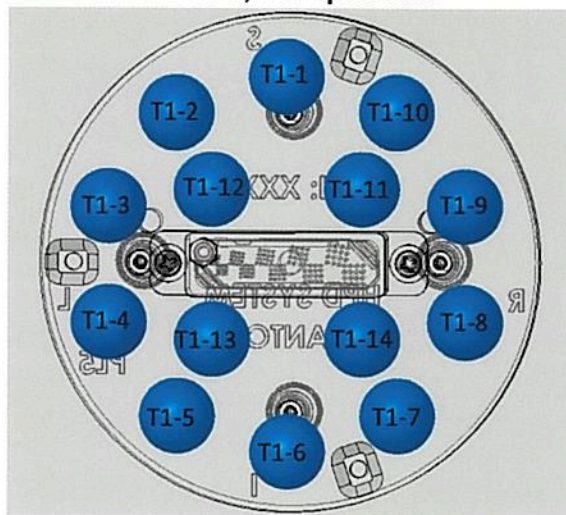


Plate 4, T2 Spheres:

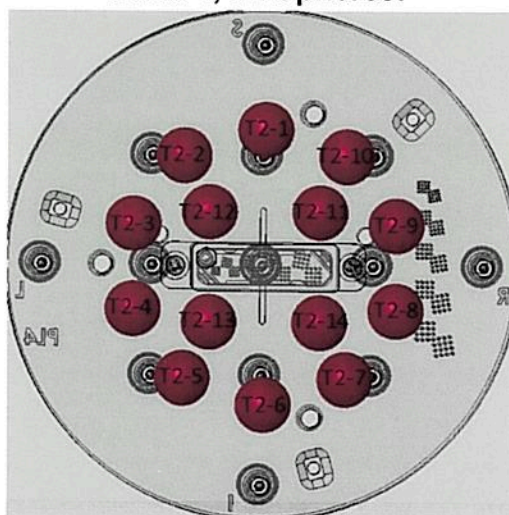
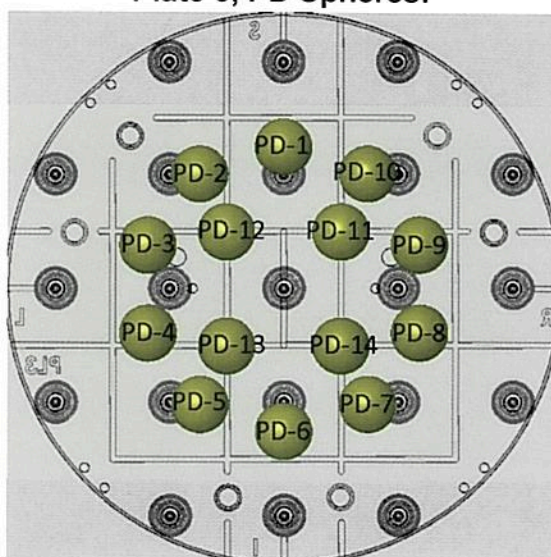


Plate 3, PD Spheres:



2. RESOLUTION INSETS

Plates 4 and 5, in coffin

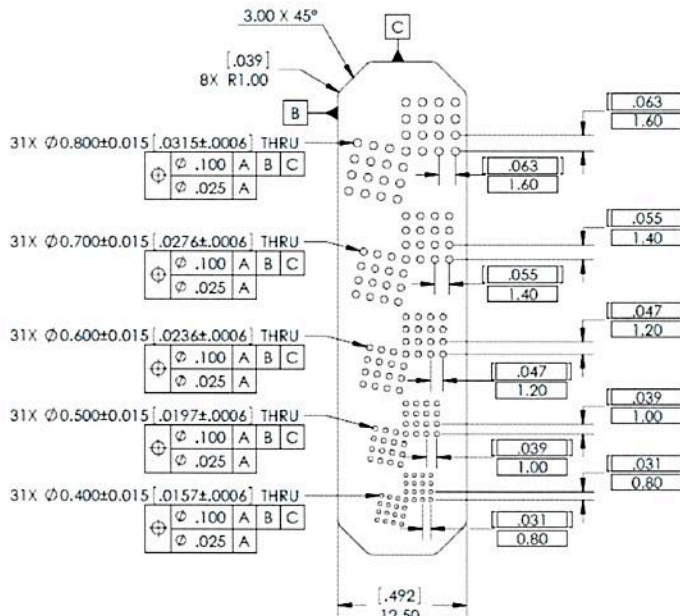
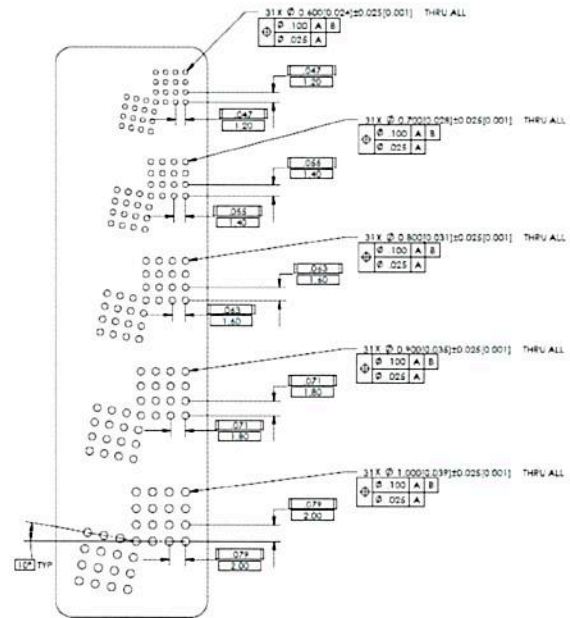
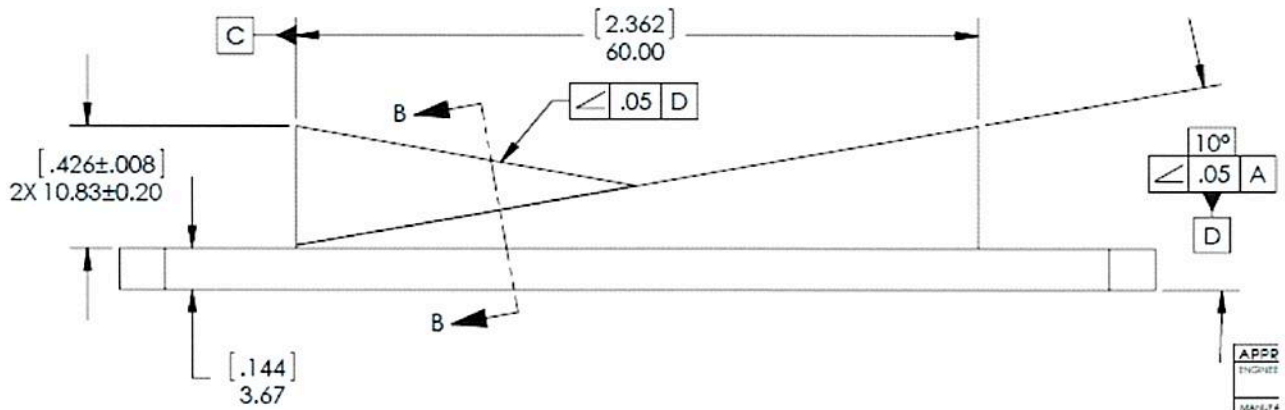


Plate 4



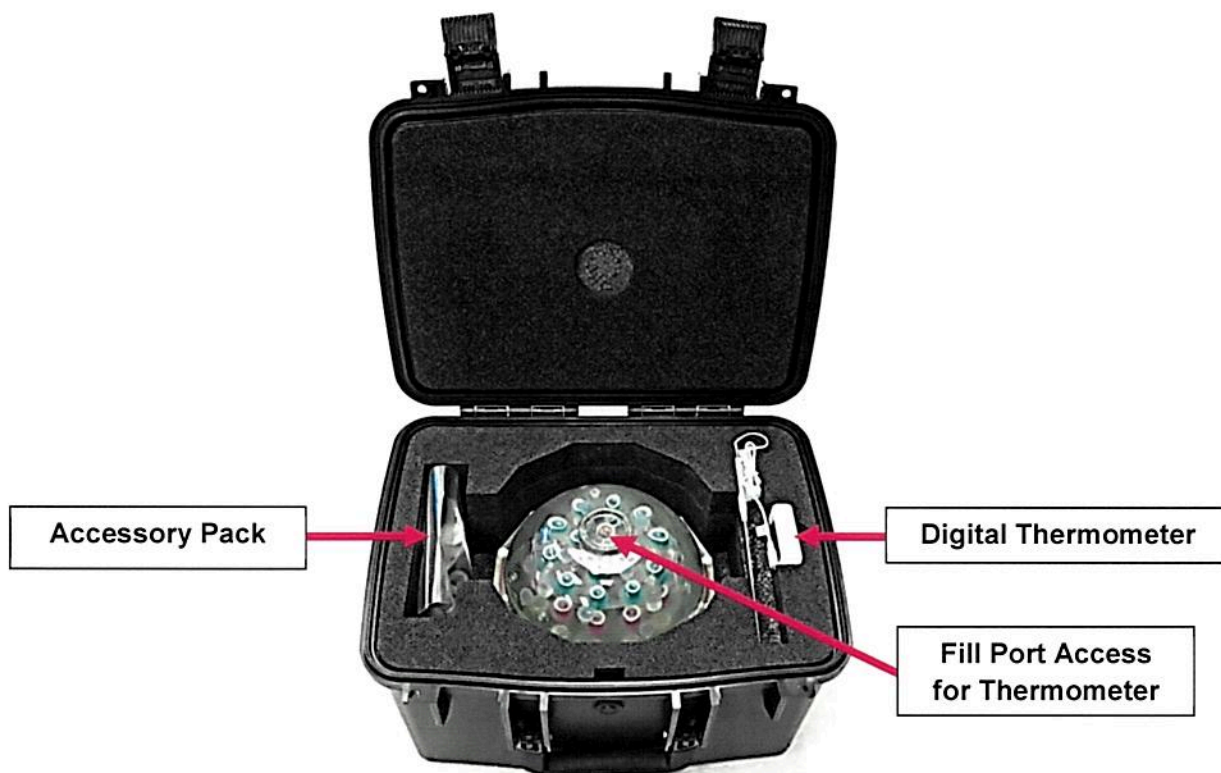
3. SLICE PROFILE WEDGES



III. SYSTEM STANDARD IMAGING

* **Safety issues:** The System Standard outer fill is DI water and is nontoxic, in case of a spill the outer fluid can be disposed down the sink. The contrast elements with paramagnetic salts are double sealed in robust polypropylene spheres, they should not leak or break even if the System Standard is dropped. There are no magnetic or metallic parts to the System Standard and it is MRI safe.*

1. Unpack the System Standard from the shipping box. Inspect for damage. If there are breaks/leaks please email support@qalibre-md.com for instructions. It is possible that small undetectable bubbles coalesced into larger bubbles during shipping. Using the pipette provided in the accessory pack, top off the System Standard using deionized or distilled water.



2. It is best if the System Standard sits in the scanner room for several hours to allow the temperature to stabilize, record the temperature using the thermometer in the System Standard box before scanning. NOTE: The thermometer has some magnetic components; please measure the System Standard temperature outside of the magnet room. T_1 , T_2 values require temperature dependent corrections to obtain ultimate accuracy of $\pm 2\%$.
3. Replace any water lost during temperature measurement with DI or distilled water using the pipette provided in the accessory pack.

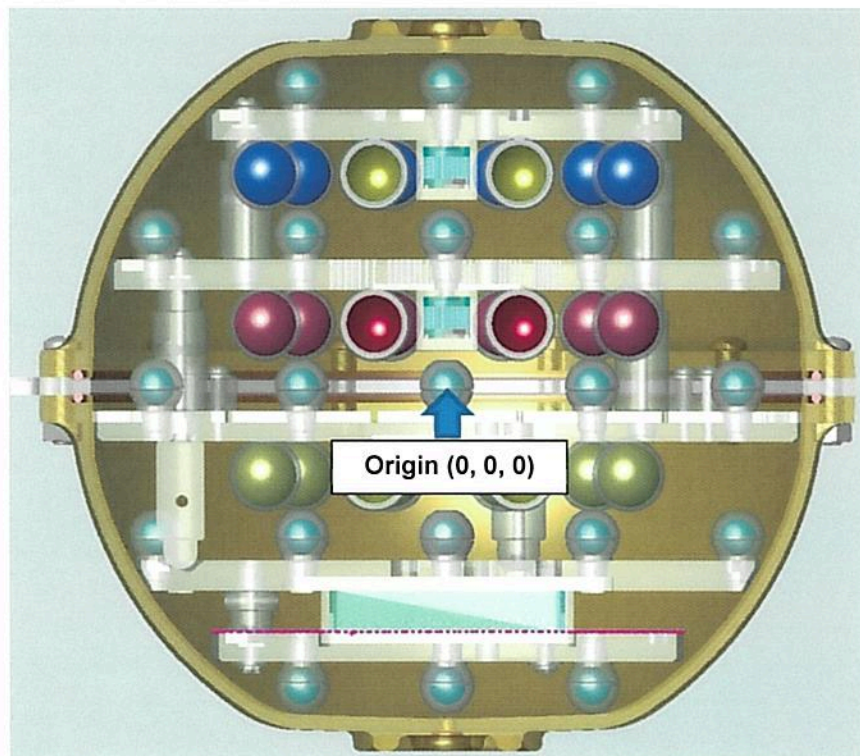
4. Load System Standard in scanner with the notch centered and at the inferior position. The plates should be oriented horizontally (parallel to the bed) with the plate markings of "S/I" and "R/L" in the appropriate places for how a patient would be oriented. A picture of a previous version of the System Standard loaded into a head coils is below.



IV. PROTOCOLS

The following pages of this document will outline the protocols and slice positions established by the International Society for Magnetic Resonance in Medicine Ad hoc Committee on Standards for Quantitative Magnetic Resonance (ISMRM SQMR) for three Vendors (General Electric, Philips, Siemens). The aim was to maintain identical scanning parameters to the best of each scanner's capabilities.

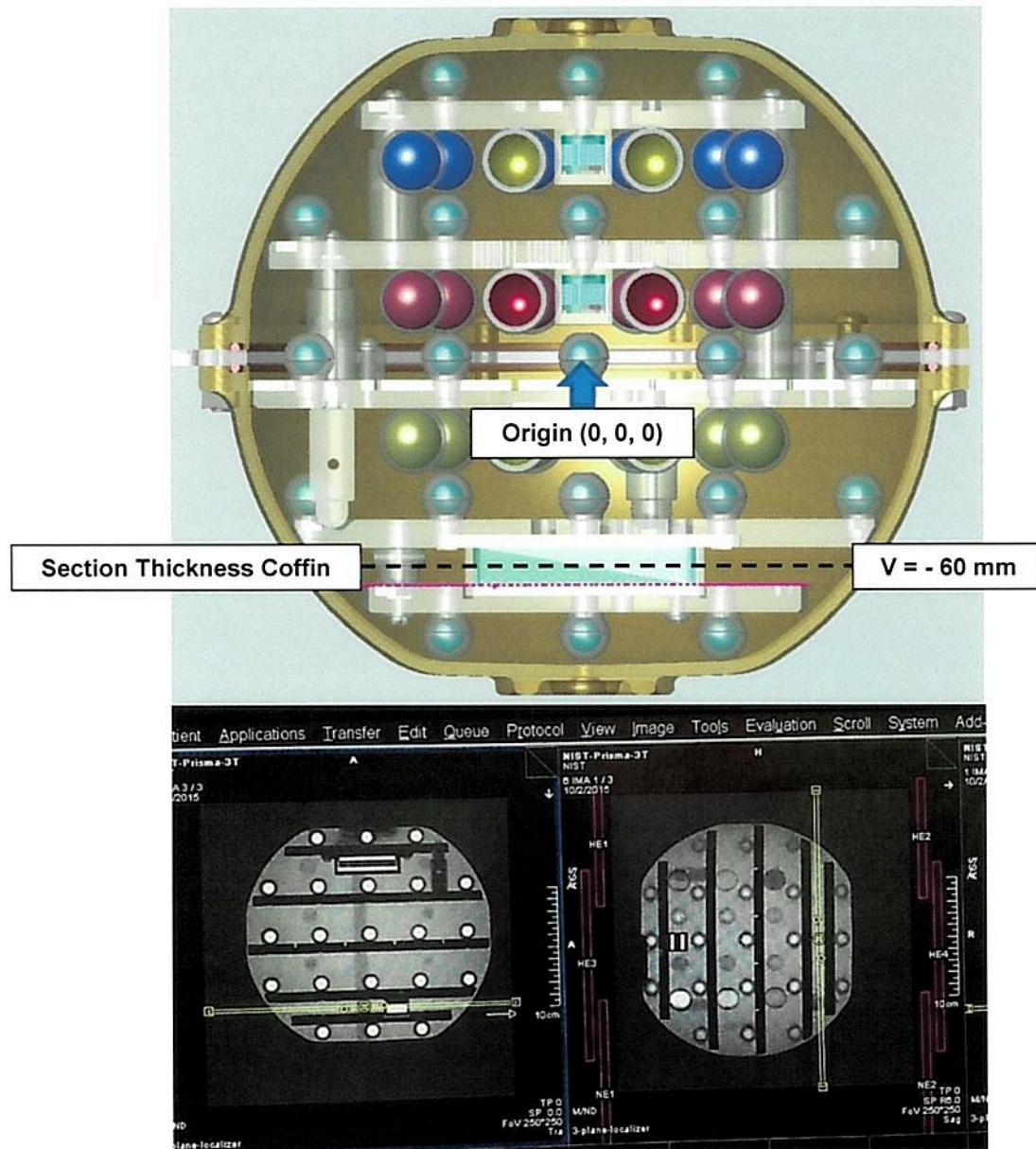
1. Isotropic Volume Sequence:



Isotropic Volume Series	GE	Philips	Siemens
Sequence	3D/SPGR	3D/SPGR	3D/RF spoiled GRE
Scan Plane	COR	COR	COR
Scan Options	EDR	3D FFE; Fast = none	
Section Thickness / Gap (mm)	1	1	1
TR (ms)	5.9	10.0	6.3
TE (ms)	1.4 (minimum)	4.0	1.9
TI Values (ms)			
Flip Angle (deg)	10	10	10
ETL	1	1	1
Number of Averages	1	1	1
Matrix (FE)	256	256	256
Matrix (PE)	256	256	256
Matrix (SE)	256	256	256
Pixel Bandwidth (Hz)	488	434	650
Bandwidth (kHz) - GE	62.5		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	250	250	250
Pixel Size (mm x mm)	0.98 x 0.98	0.98 x 0.98 x 0.98	0.98 x 0.98
PE Direction	RL	RL	RL
Notes	Resolutions are important for each of these protocols. Keep resolution and adjust TE/TR as necessary. Repeat for each plane (cor, sag, ax).	Resolutions are important for each of these protocols. Keep resolution and adjust TE/TR as necessary.	Resolutions are important for each of these protocols. Keep resolution and adjust TE/TR as necessary.
Series	Isotropic Volume	Isotropic Volume	Isotropic Volume
Approx Acq Time per Setting (min)	6.4	11.0	6.53
# Settings	1	1	1
Total Time for this Series (min)	6.4	11.0	6.53

2. Section Thickness Sequence:

In the System Standard, the section thickness inset will appear between plates 1 & 2. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.

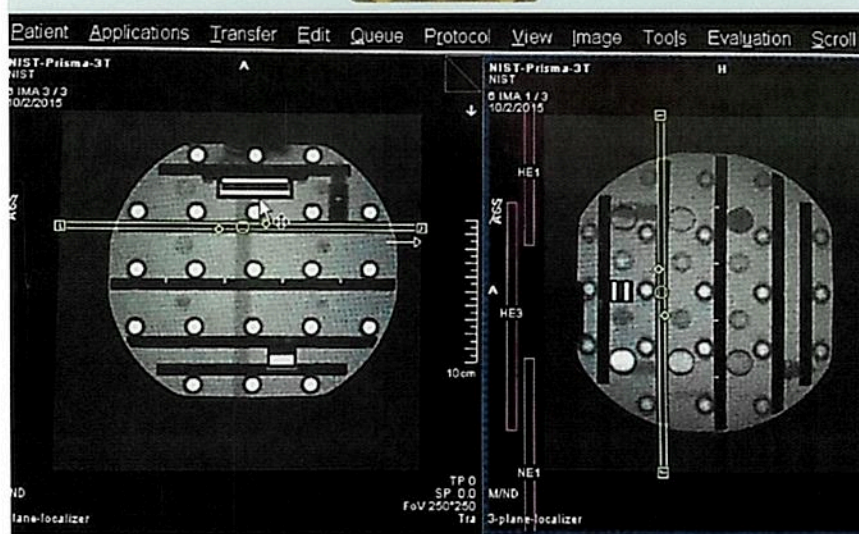
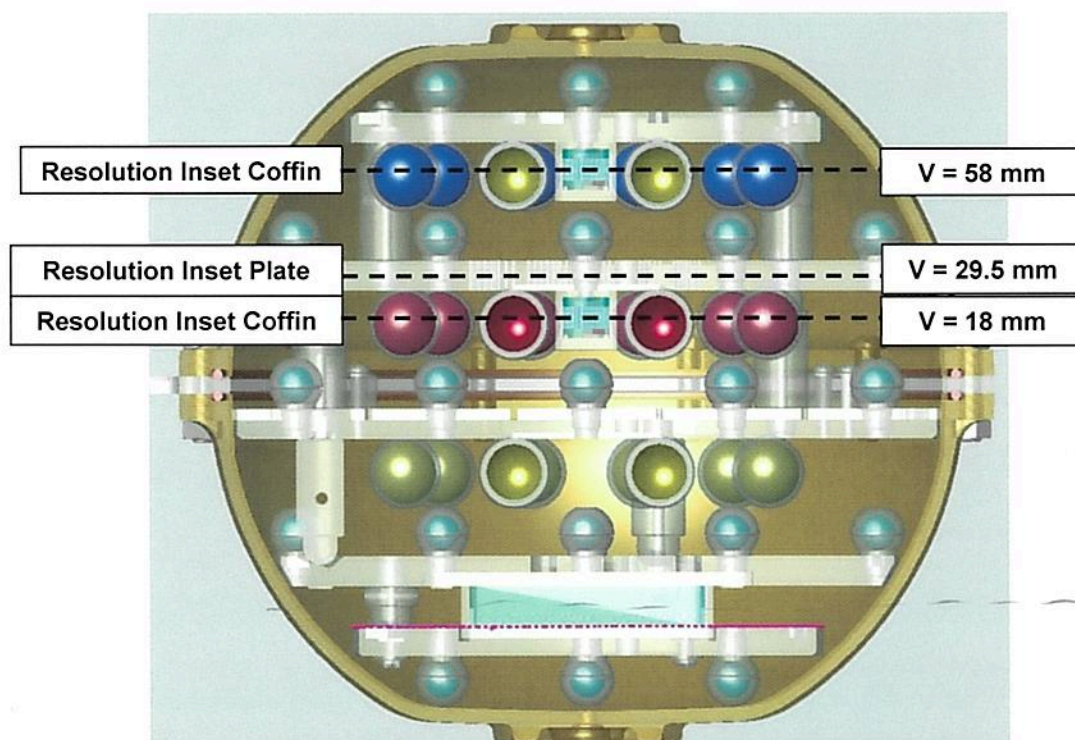


System Standard Model 130

Section Thickness Series	GE	Philips	Siemens
Sequence	2D/FSE	2D/SE	2D/TSE
Scan Plane	COR	COR	COR
Scan Options	EDR	MS SE; Fast=TSE	
Section Thickness / Gap (mm)	3/3, 5/5	3/3, 5/5	3/3, 5/5
TR (ms)	5000	2000	5000
TE (ms)	60	36	63 <i>62.</i>
TI Values (ms)			
Flip Angle (deg)		90	
ETL	8	8	8
Number of Averages	2	1	2
Matrix (FE)	512	512	512
Matrix (PE)	256	256	256
Matrix (SE)	1	1	1
Pixel Bandwidth (Hz)	244	394	184
Bandwidth (kHz) - GE	62.5		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	250	250	250
Pixel Size (mm x mm)	0.49 x 0.98	0.49 x 0.98	0.49 x 0.98
PE Direction	RL	RL	RL
Notes		Two series: one at 3mm slice and one at 5mm slice	
Series	Section Thickness	Section Thickness	Section Thickness
Approx Acq Time per Setting (min)	5.3	1.5	2.47 <i>5:25</i>
# Settings	2	2	2
Total Time for this Series (min)	10.6	3.0	4.94 <i>10:50</i>

3. Resolution Inset sequence:

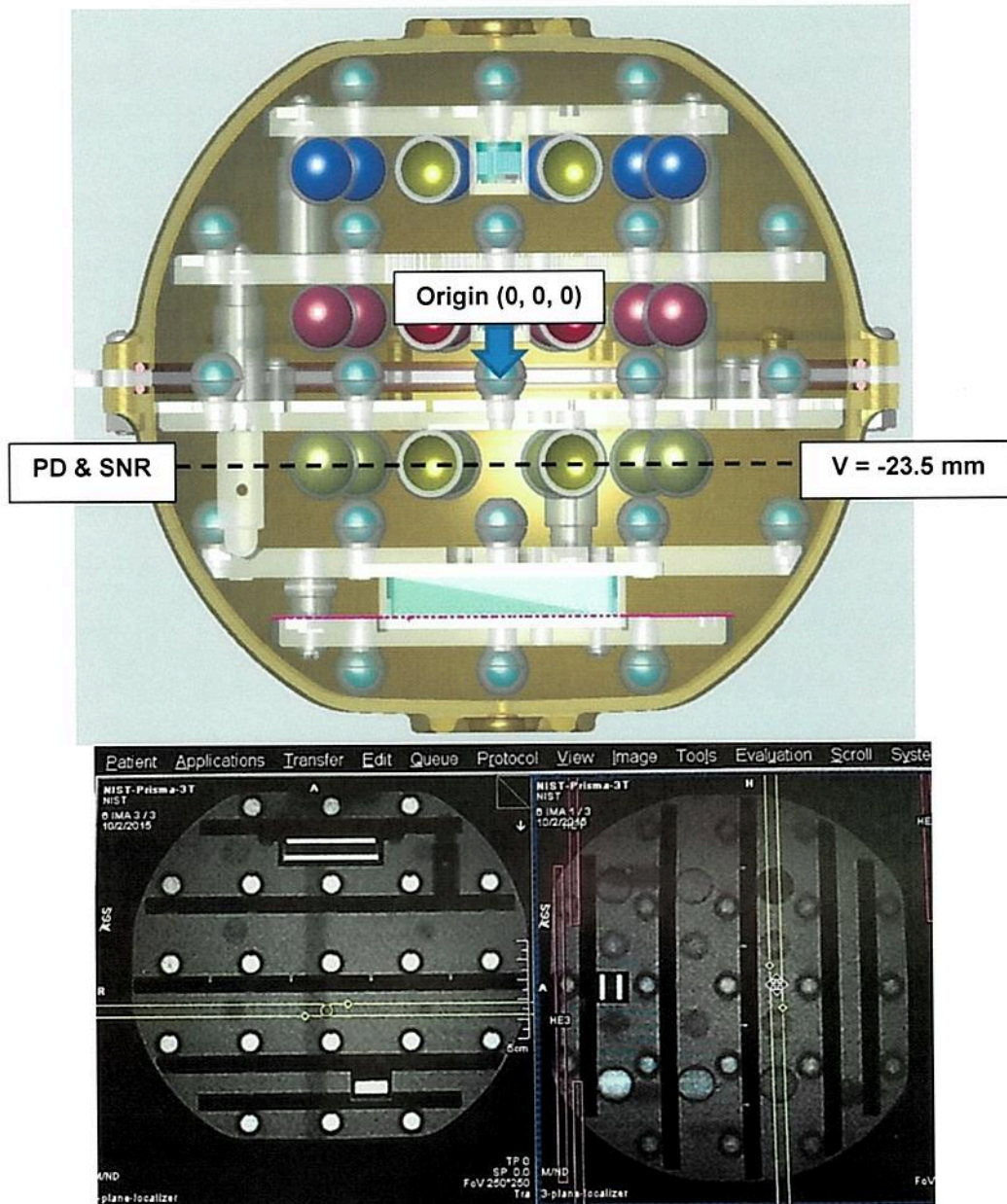
In the System Standard, the resolution inset will appear in two places. A large array ranging from 0.6 to 1.0 mm in plate 4, and from 0.4 to 1.0 in a coffin on the T1 array (green spheres in System Standard) on the bottom of plate 5. Note, that the center of the resolution inset in coffins is slightly higher than the center of the contrast spheres. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.



Resolution Inset Series	GE	Philips	Siemens
Sequence	2D/FSE	2D/SE	2D/TSE
Scan Plane	COR	COR	COR
Scan Options	EDR	MS SE; Fast=TSE	
Section Thickness / Gap (mm)	4	4	4
TR (ms)	5000	5000	5000
TE (ms)	60	60	63
TI Values (ms)			
Flip Angle (deg)		90	
ETL	8	8	8 <i>Turbo factor</i>
Number of Averages	2	2	2
Matrix (FE)	512	512	512
Matrix (PE)	512	504	512
Matrix (SE)	1	1	1
Pixel Bandwidth (Hz)	244	394	227
Bandwidth (kHz) - GE	62.5		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	250	250	250
Pixel Size (mm x mm)	0.49 x 0.49	0.49 x 0.5	0.49 x 0.49
PE Direction	RL	RL	RL
Notes	Acquire 3 sections in one acquisition, spaced appropriately to cover the three HCR inserts (if extra coffins exist). Acquire both plate and coffin insets.	Acquire 3 sections in one acquisition, spaced appropriately to cover the three HCR inserts (if extra coffins exist). Acquire both plate and coffin insets.	Acquire 3 sections in one acquisition, spaced appropriately to cover the three HCR inserts (if extra coffins exist). Acquire both plate and coffin insets.
Series	Resolution Inset	Resolution Inset	Resolution Inset
Approx Acq Time per Setting (min)	10.7	10.6	5.27 <i>10:45</i>
# Settings	2	2	2
Total Time for this Series (min)	21.4	21.2	10.54 <i>21:30</i>

4. Proton Density & SNR sequence:

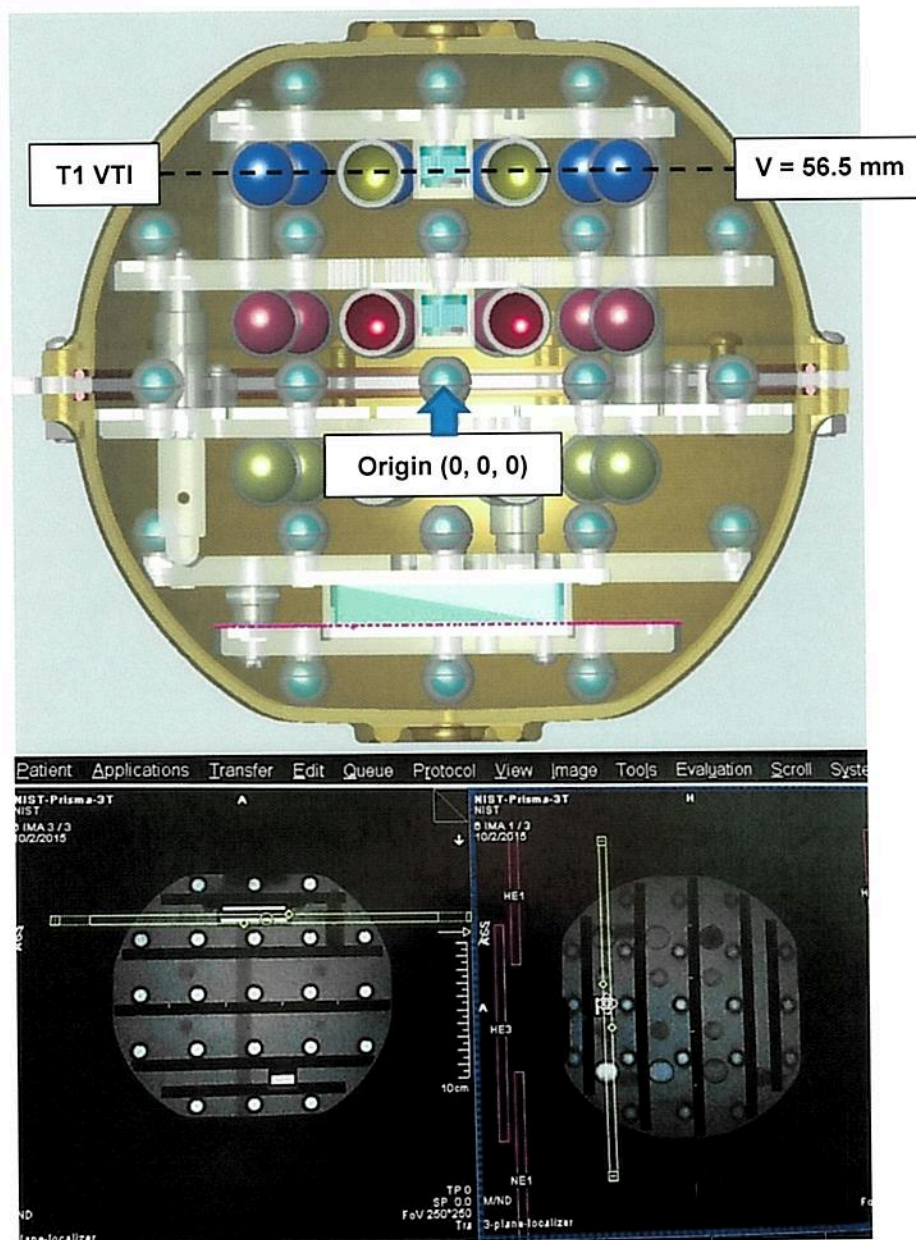
In the System Standard, the PD array will appear as spheres of yellow color. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.



PD & SNR Series	GE	Philips	Siemens
Sequence	2D/SE	2D/SE	2D/SE
Scan Plane	COR	COR	COR
Scan Options	EDR	MS SE; Fast=none	
Section Thickness / Gap (mm)	6	6	6
TR (ms)	5000	5000	5000
TE (ms)	10	10	10
TI Values (ms)			
Flip Angle (deg)		90	
ETL	1	1	1
Number of Averages	1	1	1
Matrix (FE)	256	256	256
Matrix (PE)	192	192	192
Matrix (SE)	1	1	1
Pixel Bandwidth (Hz)	244	213	153
Bandwidth (kHz) - GE	31.2		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	200 (0.8 PFOV)	250	250
Pixel Size (mm x mm)	0.98 x 0.98	0.98 x 1.3	0.98 x 0.98 <i>Recon</i>
PE Direction	RL	RL	RL
Notes	Acquire twice and use NEMA MS 1-2008 Method 1 for SNR measurement.	Acquire as single-series, but two dynamics.	Acquire twice and use NEMA MS 1-2008 Method 1 for SNR measurement.
Series	PD & SNR	PD & SNR	PD & SNR
Approx Acq Time per Setting (min)	13.8	16.0	16.07
# Settings	2	2	2
Total Time for this Series (min)	27.7	32.0	32.14

5. T1-VTI Sequence:

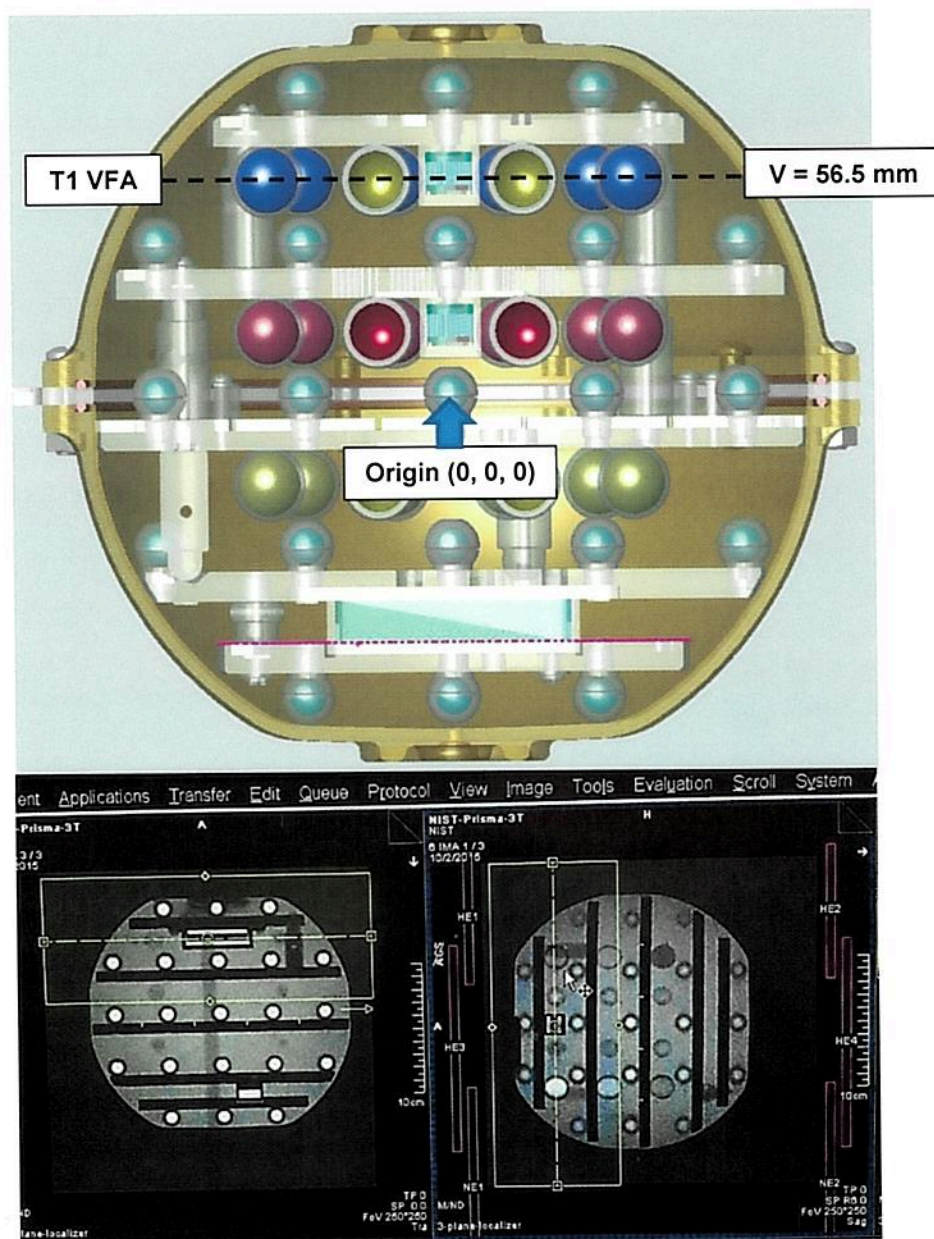
In the System Standard, the T1 array will appear as spheres of green to green/blue color. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.



T1 - VTI Series	GE	Philips	Siemens
Sequence	2D/FSE-IR	2D/IR-SK	2D/TSE-IR
Scan Plane	COR	COR	COR
Scan Options	EDR	2D IR; Fast=TSE	
Section Thickness / Gap (mm)	6	6	6
TR (ms)	4500	4500	4500
TE (ms)	Min Full (7.6)	7	6.9
TI Values (ms)	50, 75, 100, 125, 150, 250, 500, 1000, 2000, 3000	35, 75, 100, 125, 150, 250, 1000, 1500, 2000, 3000	35, 75, 100, 125, 150, 250, 1000, 1500, 2000, 3000
Flip Angle (deg)			
ETL	3	6	6
Number of Averages	1	1	1
Matrix (FE)	256	256	256
Matrix (PE)	192	252	192
Matrix (SE)	1	1	1
Pixel Bandwidth (Hz)	391	436	279
Bandwidth (kHz) - GE	50		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	200 (0.8 PFOV)	250	250
Pixel Size (mm x mm)	0.98 x 0.98	0.98 x 0.98	0.98 x 0.98
PE Direction	RL	RL	RL
Notes	Min TI allowed: 50ms APS using 50 ms TI	Reconstruct Magnitude, Real, Imaginary images. Do "FullPrep" for each of these ten series.	Use SOS (sum-of-squares) multi-channel coil reconstruction, not default ACC (adaptive coil combine). Known "line" artifacts result when using the default ACC method.
Series	T1 VTI	T1 VTI	T1 VTI
Approx Acq Time per Setting (min)	4.0	3.5	3.02 <i>2:29</i>
# Settings	11	10	10
Total Time for this Series (min)	44.0	35.0	30:20 <i>24.00</i>

6. T1-VFA Sequence:

In the System Standard, the T1 array will appear as spheres of green to green/blue color. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.

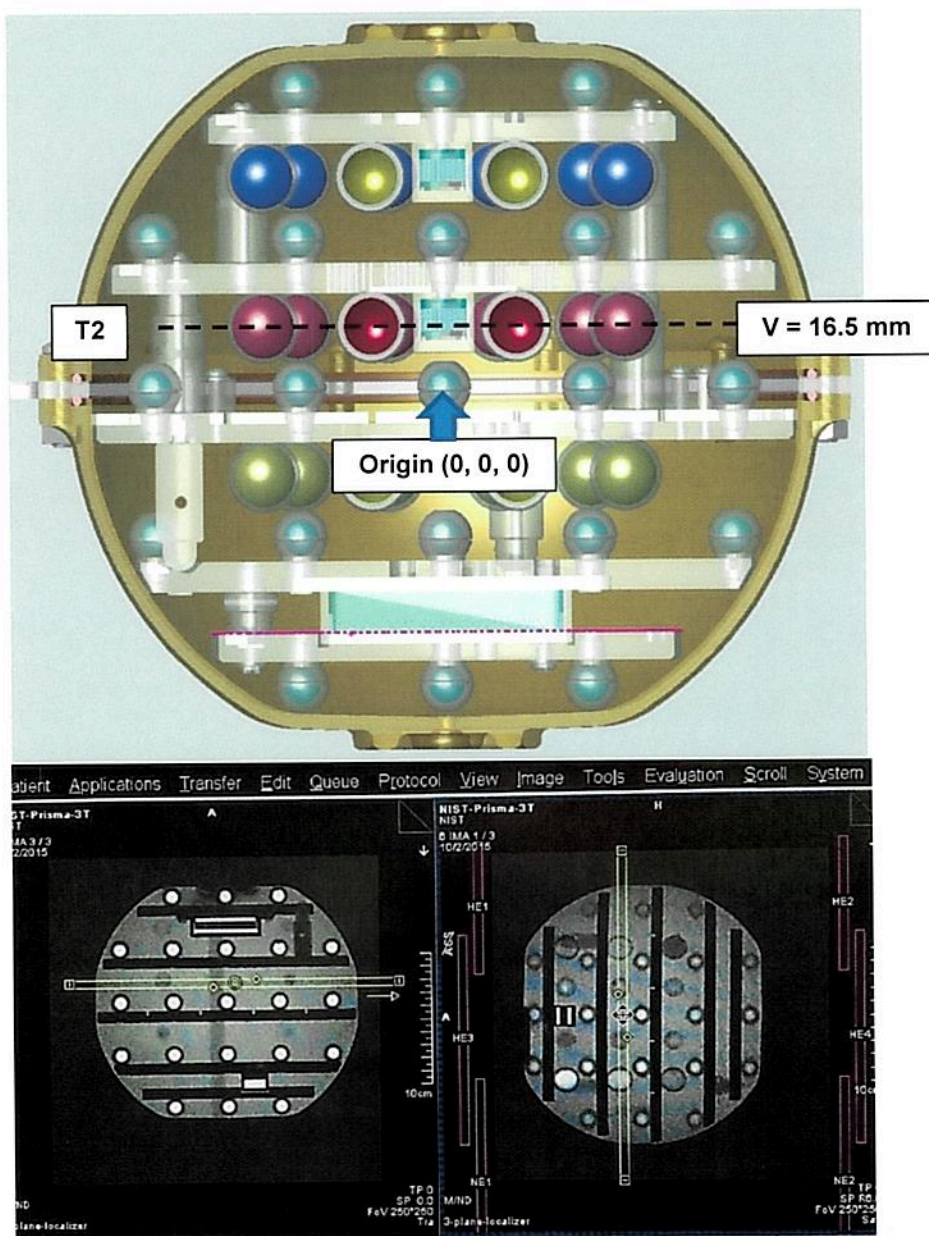


System Standard Model 130

T1 - VFA Series	GE	Philips	Siemens
Sequence	3D/FSPGR	3D/SPGR	3D/RF spoiled GRE
Scan Plane	COR	COR	COR
Scan Options	EDR/Z2	3D FFE; Fast=none	
Section Thickness / Gap (mm)	3/0	6/0	6/0 (or 3/0 if error)
TR (ms)	Min (6.0)	6.6	6.6
TE (ms)	Min (1.4)	1.8	2.44
TI Values (ms)			
Flip Angle (deg)	2, 5, 10, 20, 25, 30	2, 5, 10, 20, 25, 30	2, 5, 10, 20, 25, 30
ETL	1	1	1
Number of Averages	4	4	4
Matrix (FE)	256	256	256
Matrix (PE)	192	192	192
Matrix (SE)	34	28	32
Pixel Bandwidth (Hz)	488	904	280
Bandwidth (kHz) - GE	62.5		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	250	250	250
Pixel Size (mm x mm)	0.98 x 0.98	0.98 x 1.3 x 6	0.98 x 0.98
PE Direction	RL	RL	RL
Notes	1) Yields 30 3-mm sections. 2) Ensure gain settings do not vary between series, i.e., use MPS. Autoprescan using 15 deg FA. 3) Include fiducial spheres above & below T1 spheres in scan volume.	1) Reconstructed at 3 mm. 2) Ensure gain settings do not vary between series, i.e., use MPS. Autoprescan using 15 deg FA. 3) Include fiducial spheres above & below T1 spheres in scan volume.	1) Reconstructed at 3 mm. 2) Ensure gain settings do not vary between series, i.e., use MPS. Autoprescan using 15 deg FA. 3) Include fiducial spheres above & below T1 spheres in scan volume.
Series	T1 VFA	T1 VFA	T1 VFA
Approx Acq Time per Setting (min)	2.6	3.0	1.23 2:42
# Settings	7	7	7
Total Time for this Series (min)	18.2	21.0	8.61

7. T2 Sequence:

In the System Standard, the T2 array will appear as spheres of red color. "V" denotes the vertical direction of the System Standard and is measured from the Origin fiducial as described above in the "System Standard explanation" section and marked on the diagram. An image of the slice position from a Siemens console is shown below the System Standard diagram.



System Standard Model 130

T2 Series	GE	Philips	Siemens
Sequence	2D/SE	2D/SE	2D/se_mc
Scan Plane	COR	COR	COR
Scan Options	EDR	2D SE; Fast=none	
Section Thickness / Gap (mm)	6	6	6
TR (ms)	5000	5000	5000
TE (ms)	15, 30, 45, 60 25, 50, 75, 100 40, 80, 120, 160	11ms x 16 echos	10 to 320 ms by 10 ms
TI Values (ms)			
Flip Angle (deg)			
ETL	1	1	1
Number of Averages	1	1	1
Matrix (FE)	256	256	256
Matrix (PE)	192	192	192
Matrix (SE)	1	1	1
Pixel Bandwidth (Hz)	156	172	227
Bandwidth (kHz) - GE	20.0		
FOV (FE, mm)	250	250	250
FOV (PE, mm)	200 (0.8 PFOV)	250	250
Pixel Size (mm x mm)	0.98 x 0.98	0.98 x 1.3	0.98 x 0.98
PE Direction	RL	RL	RL
Notes	Ensure gain settings do not vary between series, i.e., use MPS. APS with TE=15ms.	Ensure gain settings do not vary between series, i.e., use MPS.	
Series	T2	T2	T2
Approx Acq Time per Setting (min)	13.8	16.0	16.1
# Settings	3	1	1
Total Time for this Series (min)	41.5	16.0	16.1

V. SYSTEM STANDARD MATERIAL PROPERTIES*

These values are for Serial Numbers 0001-0041. For Serial Numbers 0042 and above, refer to p. 22. *Please cite data as ISMRM Singapore Proceedings, program number 3290 when referencing these values in your publications.*

1. T₁ Contrast Spheres

1.5T

Sample Name	Nominal Concentration, NiCl ₂ (± 5%, mM)	T ₁ (ms)	T ₁ Standard Deviation, SD _{3R} (ms)	T ₂ (ms)	T ₂ Standard Deviation, SD _{3R} (ms)
T1-1	0.299	2 033	4.6	1 669	0.5
T1-2	0.623	1 489	1.4	1 244	0.6
T1-3	1.072	1 012	0.2	859.3	0.17
T1-4	1.720	730.8	1.10	628.5	0.13
T1-5	2.617	514.1	0.06	446.3	0.11
T1-6	3.912	367.9	0.66	321.2	0.30
T1-7	5.731	260.1	0.04	227.7	0.07
T1-8	8.297	184.6	0.02	161.9	0.06
T1-9	11.936	132.7	0.02	117.1	0.03
T1-10	17.070	92.7	0.09	81.9	0.02
T1-11	24.326	65.4	0.10	57.7	0.02
T1-12	34.590	46.32	0.010	41.0	0.01
T1-13	49.122	32.45	0.012	28.7	0.03
T1-14	69.680	22.859	0.043 7	20.2	0.01

3.0T

Sample Name	Nominal Concentration, NiCl ₂ (± 5%, mM)	T ₁ (ms)	T ₁ Standard Deviation, SD _{3R} (ms)	T ₂ (ms)	T ₂ Standard Deviation, SD _{3R} (ms)
T1-1	0.299	1 989	1.0	1 465	1.0
T1-2	0.623	1 454	2.5	1 076	1.8
T1-3	1.072	984.1	0.33	717.9	1.12
T1-4	1.720	706	1.5	510.1	1.36
T1-5	2.617	496.7	0.41	359.6	0.22
T1-6	3.912	351.5	0.91	255.5	0.07
T1-7	5.731	247.13	0.086	180.8	0.04
T1-8	8.297	175.3	0.11	127.3	0.14
T1-9	11.936	125.9	0.33	90.3	0.14
T1-10	17.070	89.0	0.17	64.3	0.05
T1-11	24.326	62.7	0.13	45.7	0.12
T1-12	34.590	44.53	0.090	31.86	0.02
T1-13	49.122	30.84	0.016	22.38	0.02
T1-14	69.680	21.719	0.005 4	15.83	0.03

2. T₂ Contrast Spheres

1.5T

Sample Name	Nominal Concentration, MnCl ₂ (± 5%, mM)	T ₁ (ms)	T ₁ Standard Deviation, SD _{3R} (ms)	T ₂ (ms)	T ₂ Standard Deviation, SD _{3R} (ms)
T2-1	0.013	2 376	2.2	939.4	1.11
T2-2	0.021	2 183	1.2	594.3	0.32
T2-3	0.031	1 870	6.0	416.5	1.13
T2-4	0.047	1 539	3.8	267.0	0.11
T2-5	0.069	1 237	0.4	184.9	0.11
T2-6	0.101	1 030	1.7	140.6	0.05
T2-7	0.145	752.2	1.12	91.76	0.029
T2-8	0.207	550.2	0.18	64.84	0.029
T2-9	0.296	413.4	0.29	45.28	0.029
T2-10	0.421	292.9	0.15	30.62	0.014
T2-11	0.599	194.9	0.08	19.76	0.017
T2-12	0.849	160.2	0.23	15.99	0.012
T2-13	1.104	106.4	0.02	10.47	0.006
T2-14	1.704	83.33	0.10	8.15	0.011

3.0T

Sample Name	Nominal Concentration, MnCl ₂ (± 5%, mM)	T ₁ (ms)	T ₁ Standard Deviation, SD _{3R} (ms)	T ₂ (ms)	T ₂ Standard Deviation, SD _{3R} (ms)
T2-1	0.013	2 480	10.8	581.3	0.39
T2-2	0.021	2 173	14.7	403.5	0.55
T2-3	0.031	1 907	10.3	278.1	0.28
T2-4	0.047	1 604	7.2	190.94	0.011
T2-5	0.069	1 332	0.8	133.27	0.073
T2-6	0.101	1 044	3.2	96.89	0.049
T2-7	0.145	801.7	1.70	64.07	0.034
T2-8	0.207	608.6	1.03	46.42	0.014
T2-9	0.296	458.4	0.33	31.97	0.083
T2-10	0.421	336.5	0.18	22.56	0.012
T2-11	0.599	244.2	0.09	15.813	0.006 1
T2-12	0.849	176.6	0.09	11.237	0.005 7
T2-13	1.104	126.9	0.03	7.911	0.003 7
T2-14	1.704	90.9	0.05	5.592	0.005 5

*For SN 0042 and greater the following table is referenced to NIST through ICP-MS and ICP-OES

Table 1: Table of Values for Serial Number SN: 0042 and Higher.

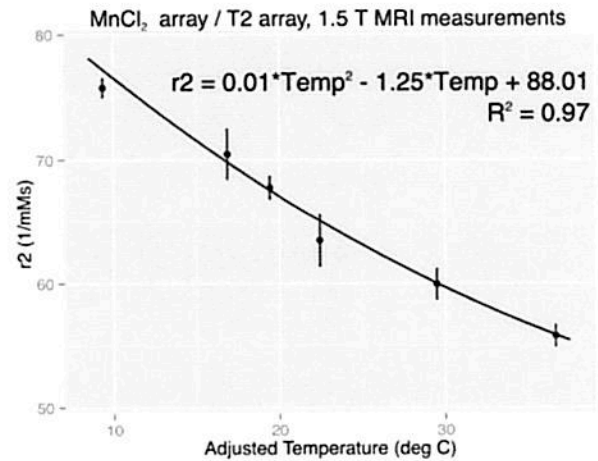
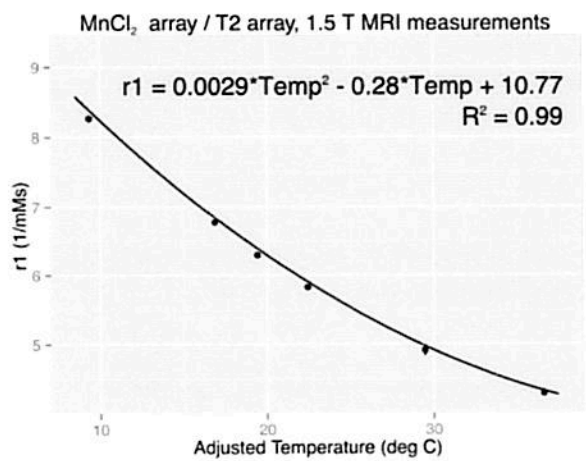
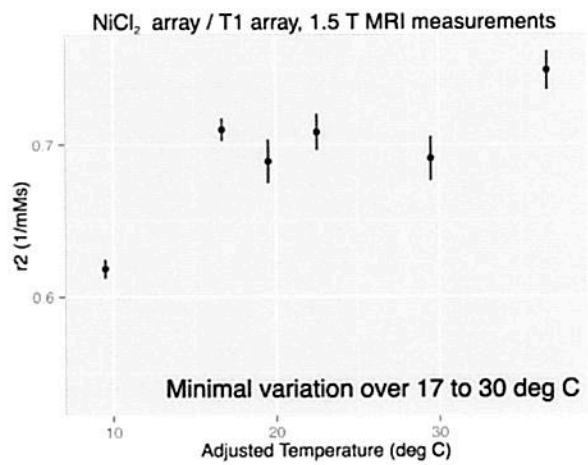
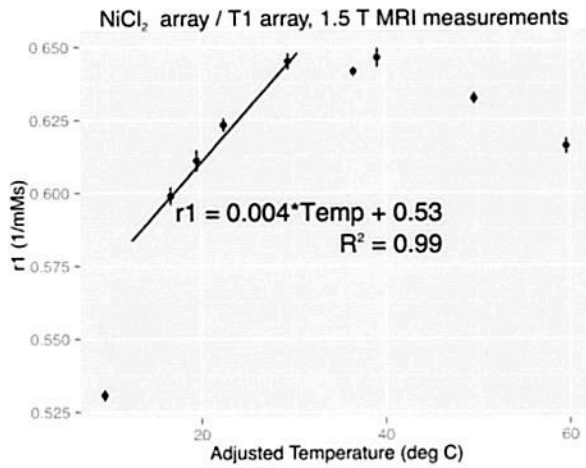
**Field Strength =
1.5T**

	T1 (ms)	T2 (ms)		T1 (ms)	T2 (ms)
T1-1	1879	1542	T2-1	2640	1044
T1-2	1432	1196	T2-2	2292	623.9
T1-3	1027	871.7	T2-3	1923	428.3
T1-4	751.3	646.1	T2-4	1489	258.4
T1-5	527.0	457.5	T2-5	1245	186.1
T1-6	384.1	335.3	T2-6	1004	137.0
T1-7	272.3	238.4	T2-7	733.9	89.52
T1-8	194.5	170.6	T2-8	533.1	62.82
T1-9	137.8	121.6	T2-9	400.3	43.84
T1-10	94.7	83.7	T2-10	261.0	27.28
T1-11	67.0	59.1	T2-11	189.8	19.24
T1-12	48.14	42.6	T2-12	154.7	15.44
T1-13	34.35	30.4	T2-13	102.1	10.05
T1-14	24.16	21.3	T2-14	79.65	7.79

**Field Strength =
3T**

	T1 (ms)	T2 (ms)		T1 (ms)	T2 (ms)
T1-1	1838	1354	T2-1	2756	645.8
T1-2	1398	1035	T2-2	2281	423.6
T1-3	998.3	728.3	T2-3	1961	286.0
T1-4	725.8	524.4	T2-4	1552	184.8
T1-5	509.1	368.6	T2-5	1341	134.1
T1-6	367.0	266.7	T2-6	1017	94.40
T1-7	258.7	189.3	T2-7	782.1	62.51
T1-8	184.7	134.1	T2-8	589.7	44.98
T1-9	130.8	93.8	T2-9	443.8	30.95
T1-10	90.9	65.7	T2-10	299.8	20.10
T1-11	64.2	46.8	T2-11	237.8	15.40
T1-12	46.28	33.11	T2-12	170.5	10.85
T1-13	32.65	23.69	T2-13	121.8	7.59
T1-14	22.95	16.73	T2-14	86.9	5.35

3. Temperature Dependence @ 1.5T



4. Proton Density Spheres

Contrast ID	Concentration (% water)	H ₂ O mass (gm)	D ₂ O mass(gm)
PD-1	5	0.746	15.758
PD-2	10	1.491	14.932
PD-3	15	2.238	14.106
PD-4	20	2.984	13.279
PD-5	25	3.731	12.451
PD-6	30	4.478	11.624
PD-7	35	5.226	10.796
PD-8	40	5.974	9.968
PD-9	50	7.470	8.310
PD-10	60	8.968	6.651
PD-11	70	10.468	4.990
PD-12	80	11.968	3.328
PD-13	90	13.470	1.665
PD-14	100	14.973	0.000